

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 24 June 2026

Project Name: Visualization Tools for Electric Vehicle Charge and Range Analysis

Technical Architecture



Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive dashboard	Tableau
2	Application Logic-1	Data preprocessing	SQL
3	Application Logic-2	Data analysis	Python
4	Application Logic-3	Dashboard creation & story	Tableau
5	Database	EV dataset storage	MySQL
6	Cloud Database	Not used	-
7	File Storage	CSV/Excel files	Local File System
8	External API-1	Not used	-
9	External API-2	Not used	-
10	Machine Learning Model	Not used	-
11	Infrastructure	Local deployment	Windows + MySQL + Tableau

Table-2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Visualization and database tools	Tableau Public, MySQL
2	Security	Password-protected local database	MySQL Authentication
3	Scalable Architecture	Supports additional EV datasets	MySQL + Tableau
4	Availability	Accessible on local machine	Windows OS
5	Performance	Optimized SQL queries and dashboards	SQL + Tableau